

Circle Packing

A Visual Introduction

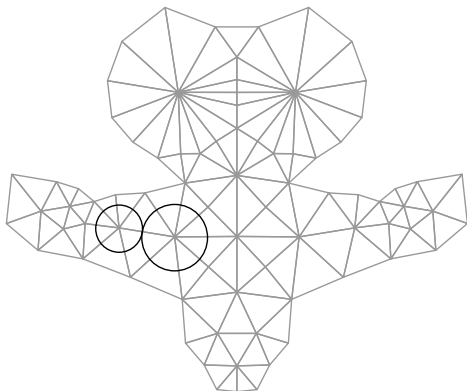
James Ashe

Mars Hill University

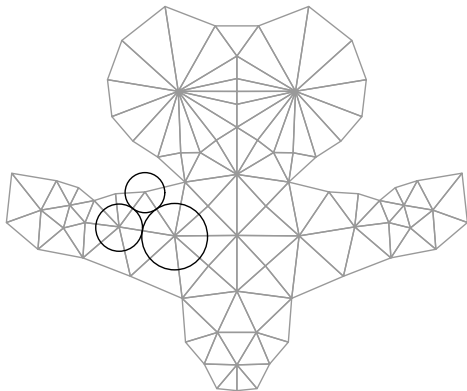


What is a Circle Packing?

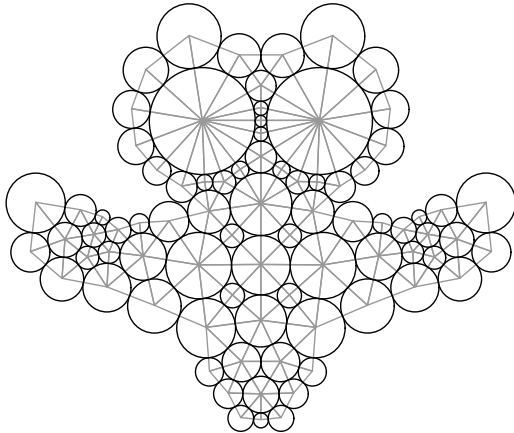
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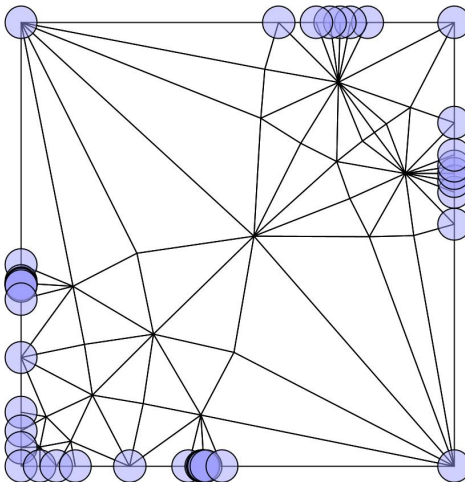


A collection of circles with a pattern such that

1. The circles have one and only one circle, C_i , associated with each vertex v_i in the pattern.
2. Any two neighbors in the circle collection are *tangent*.
3. $\{C_i, C_j, C_k\}$ form a positively oriented triple whenever $\{v_i, v_j, v_k\}$ form a positively oriented face in the pattern.

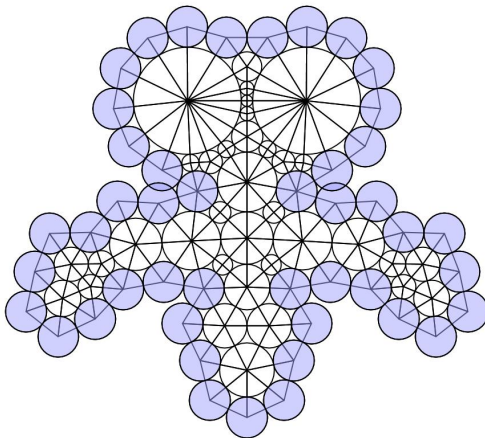
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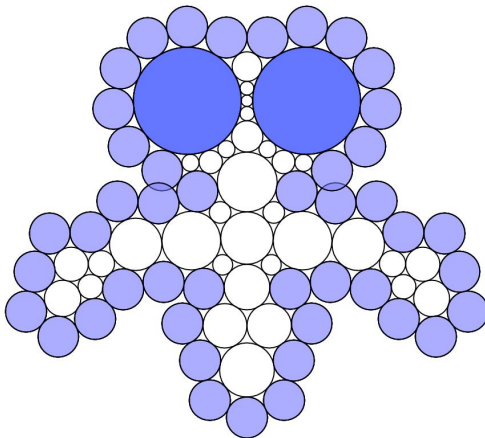
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Complex + Circles = Geometry



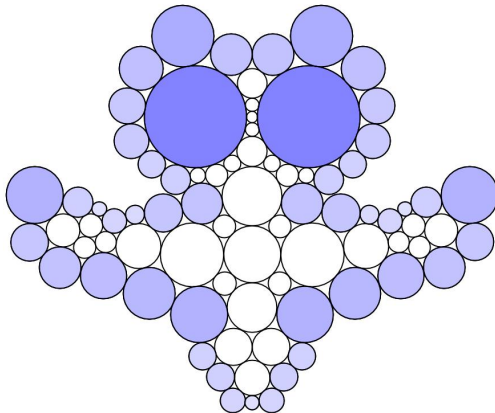
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- Border conditions introduce flexibility.

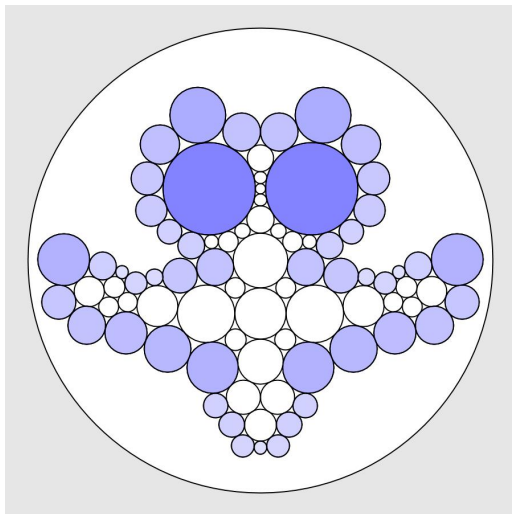
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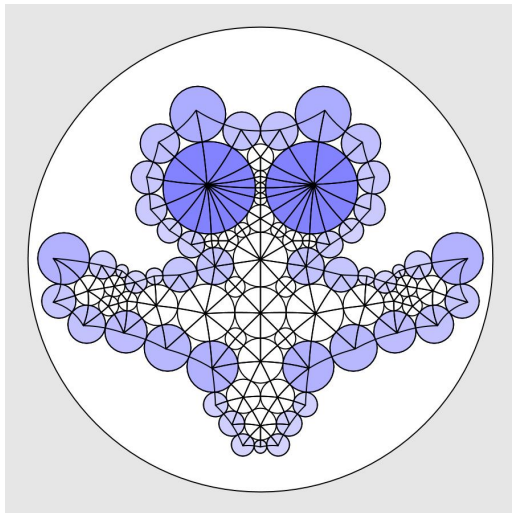
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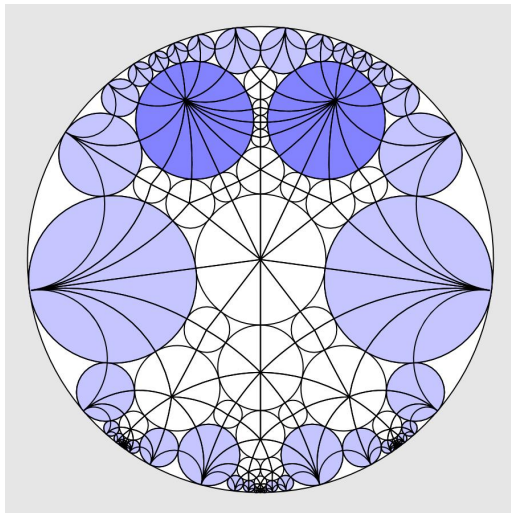
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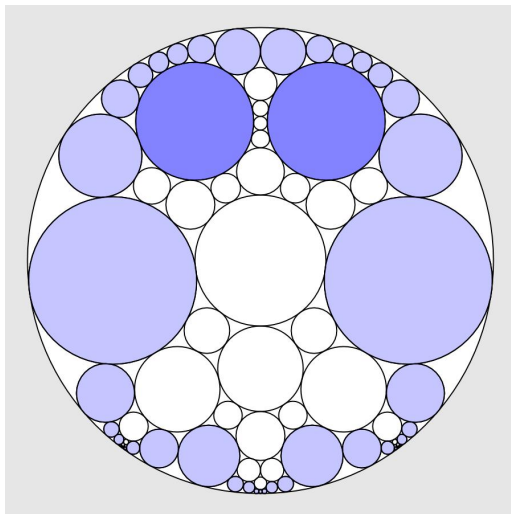
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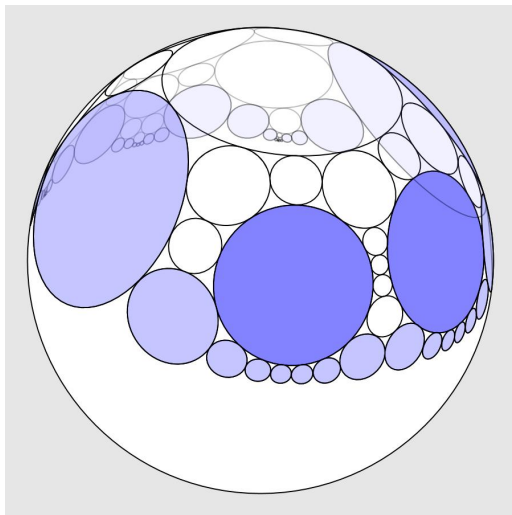
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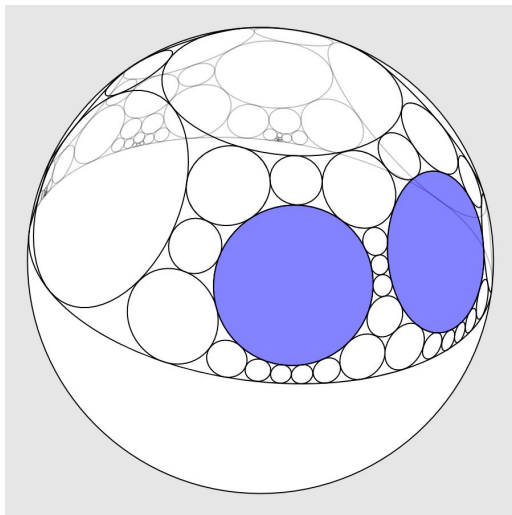
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- The sphere, $\mathbb{P}$.



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Moving between Geometries

Via Stereographic Projections of the Riemann sphere.

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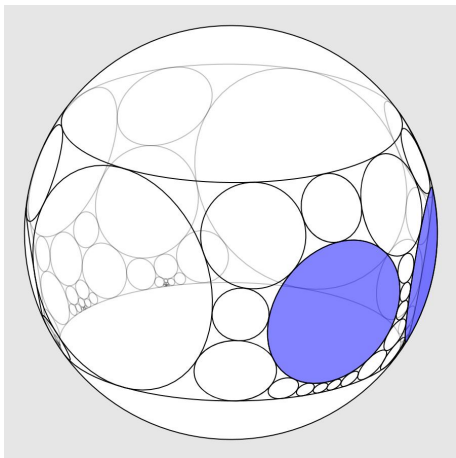
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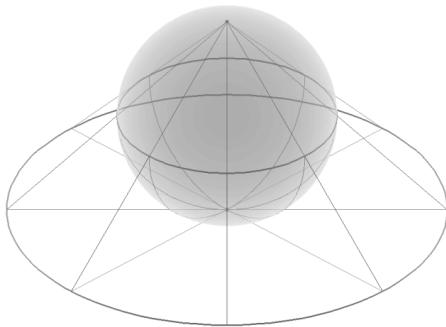
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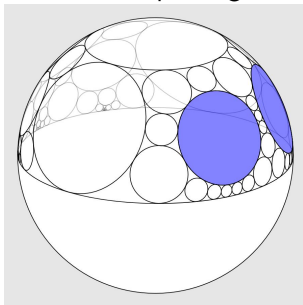
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There exists a CP on the sphere for any finite triangulation of the sphere. Furthermore, this CP is unique up to Möbius transformations.

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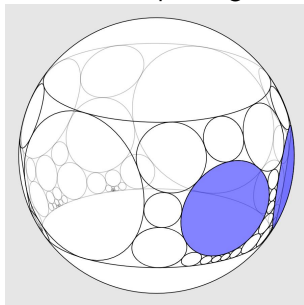
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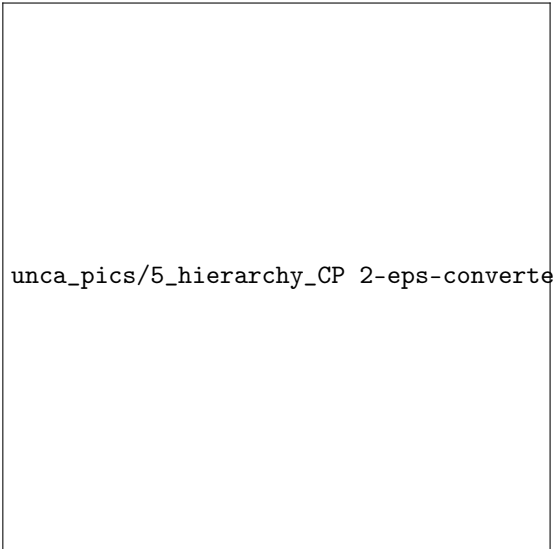


Preliminaries for KAT Proof

Hierarchy of a circle packing

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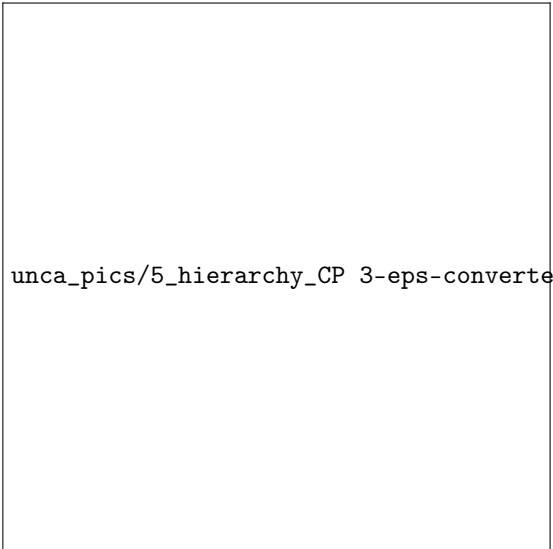
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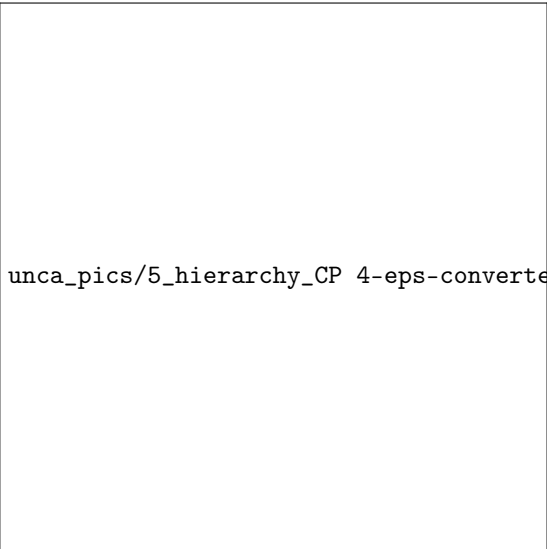
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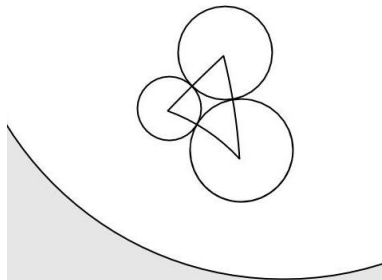
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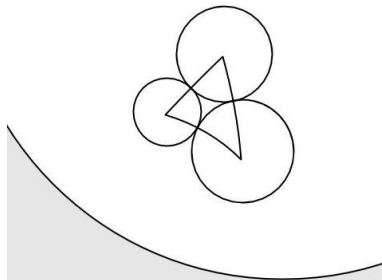


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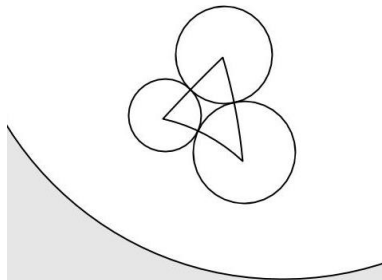


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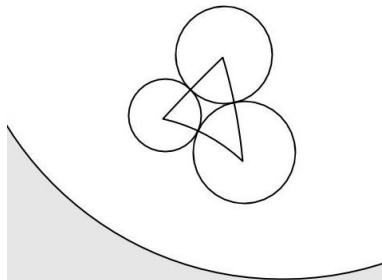


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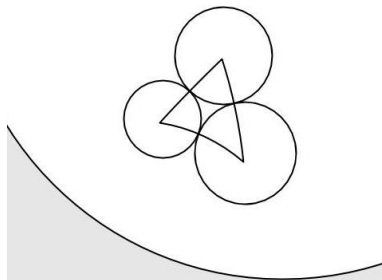


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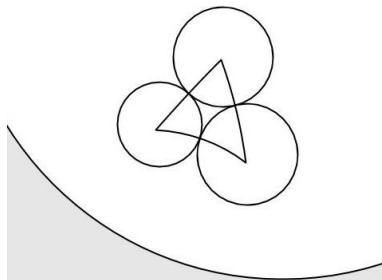


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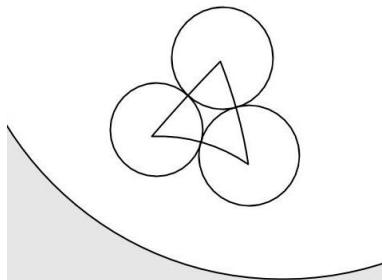


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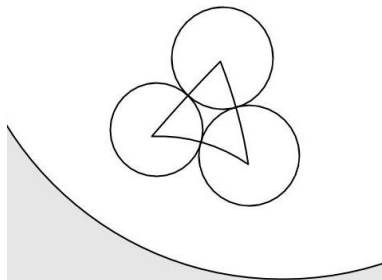


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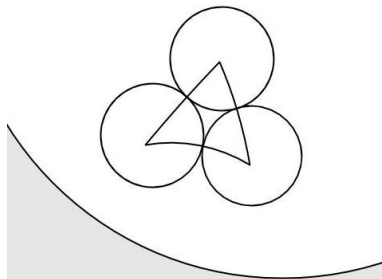


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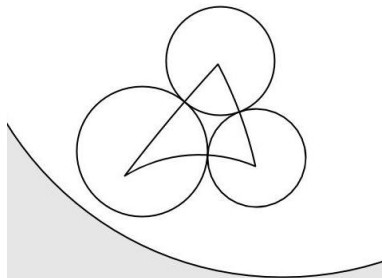


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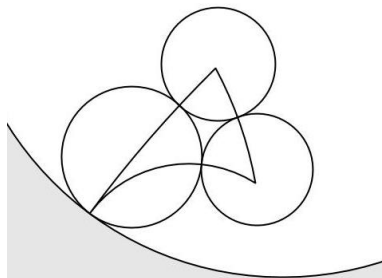


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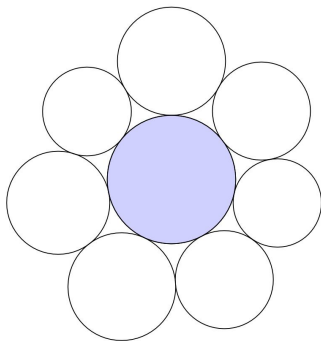
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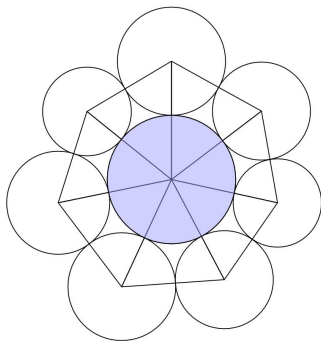
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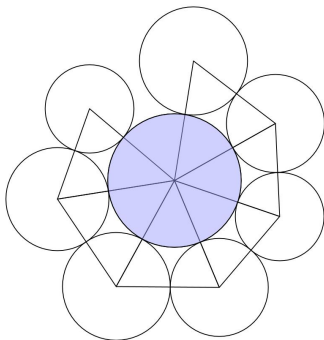


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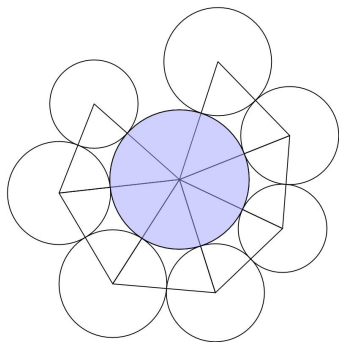


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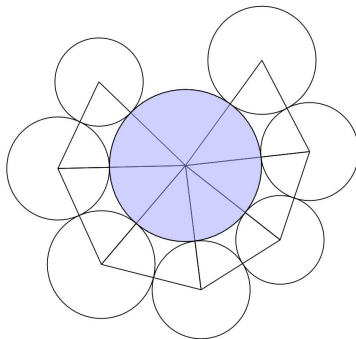


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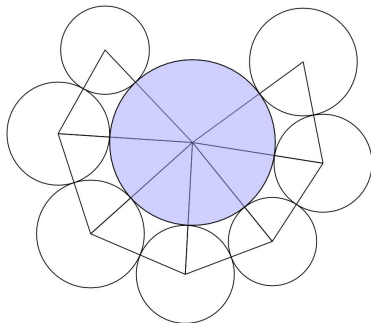


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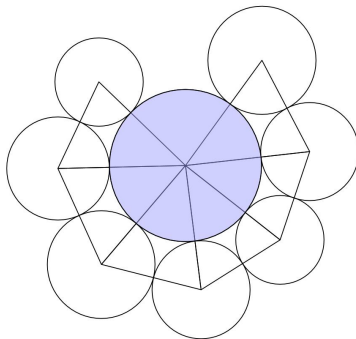


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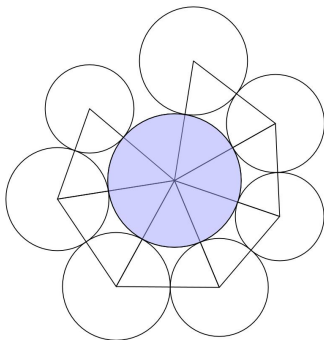


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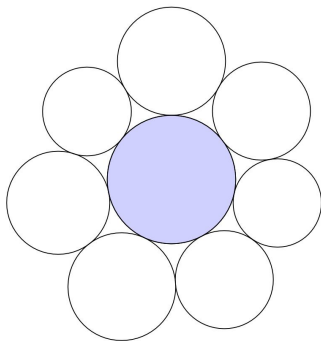
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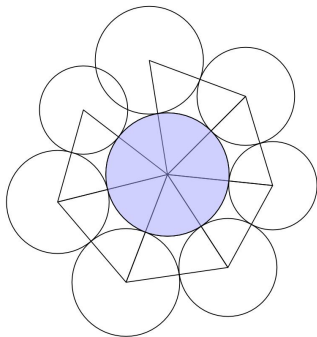
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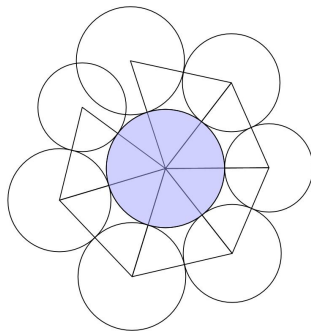
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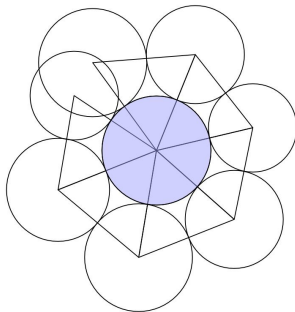
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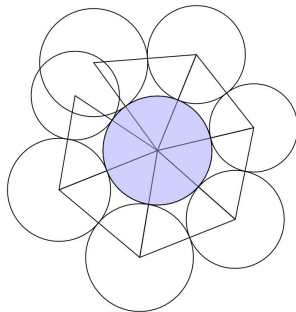
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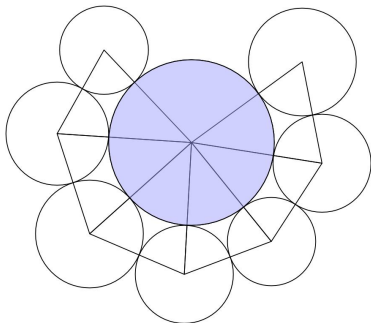
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- *Packing condition: Angle sum must be $2\pi n$.*

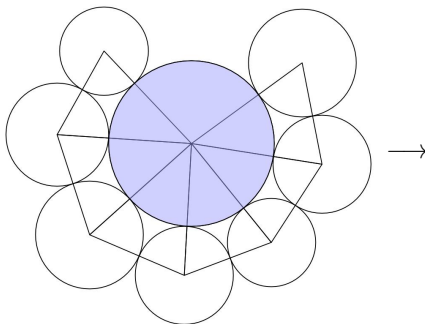
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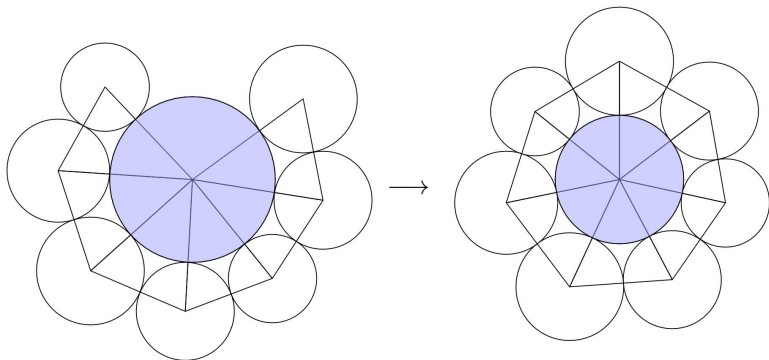
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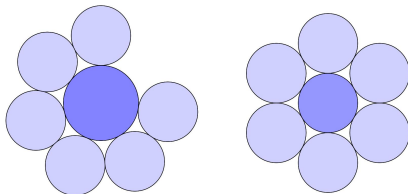
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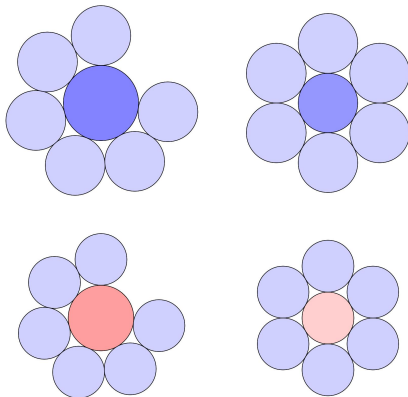
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- ...with some details.

- Maximality
argument $\Rightarrow$ uniqueness.

Theorem

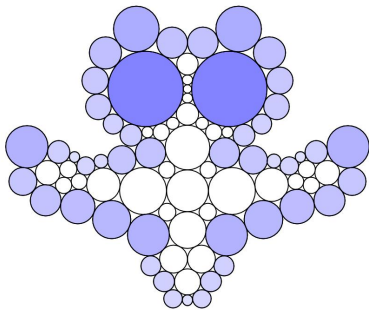
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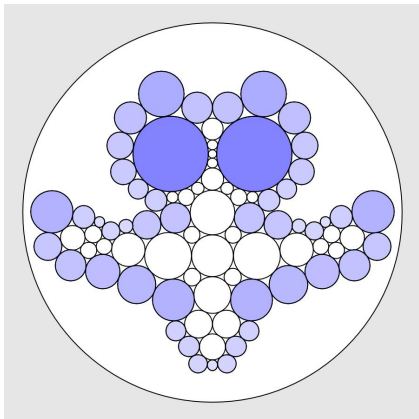
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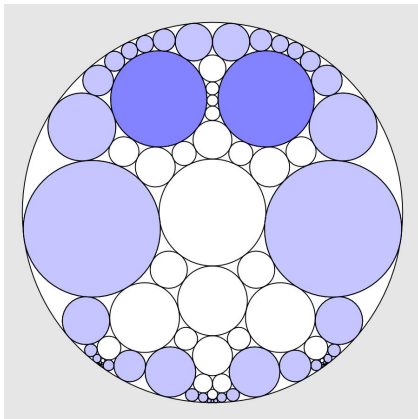
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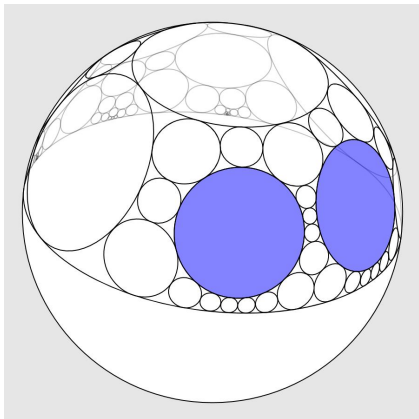
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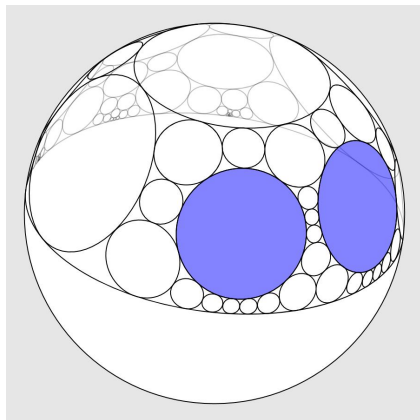
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Koebe-Andreev-Thurston Theorem (KAT)

There exists a CP on the sphere for any finite triangulation of the sphere. Furthermore, this CP is unique up to Möbius transformations.



Circle Packings as Discrete Analytical Function

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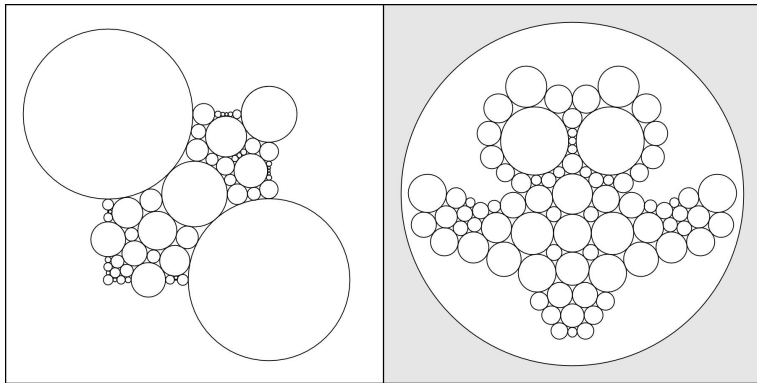
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Circle Packings as Discrete Analytical Function

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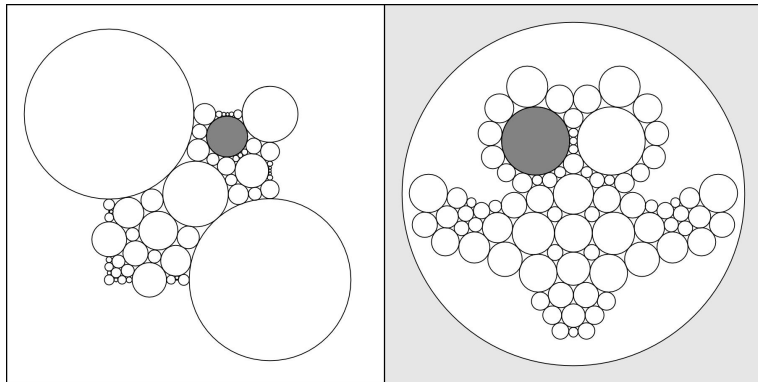
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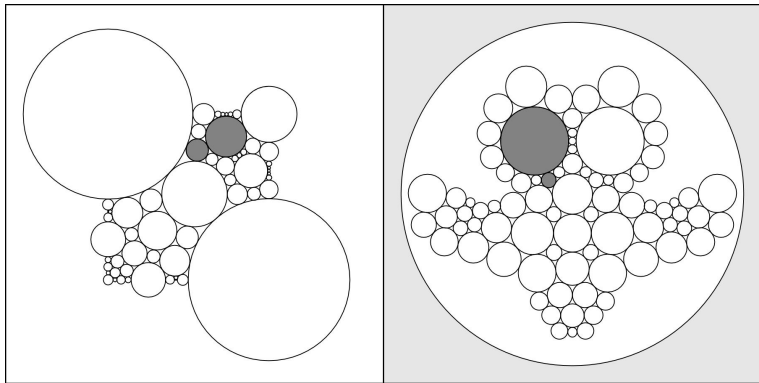
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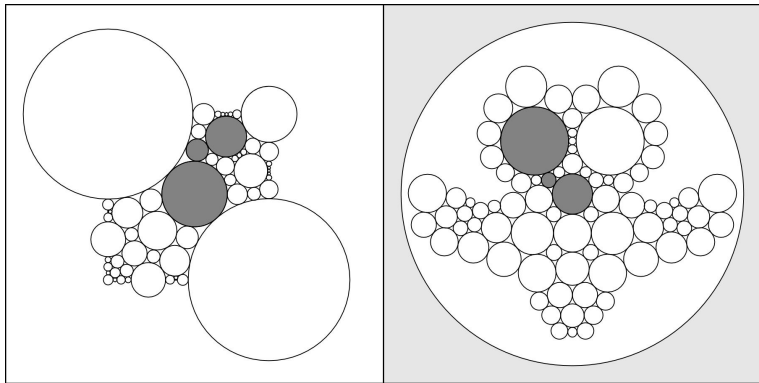
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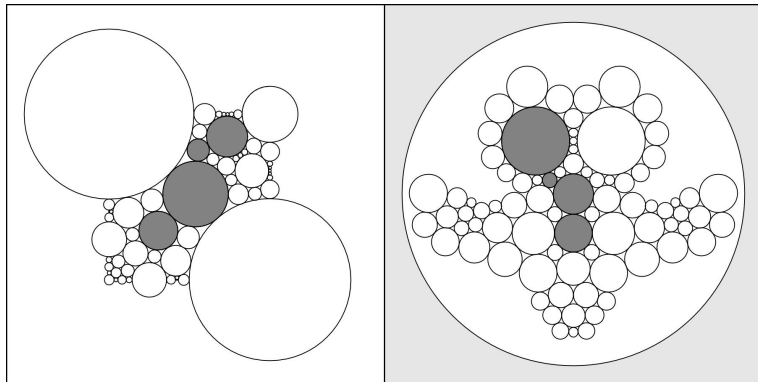
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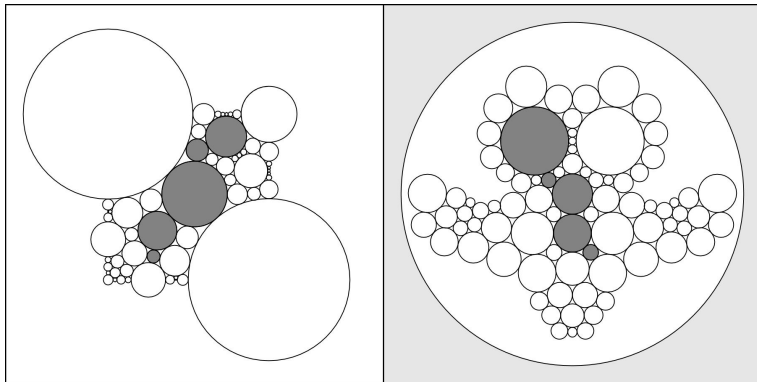
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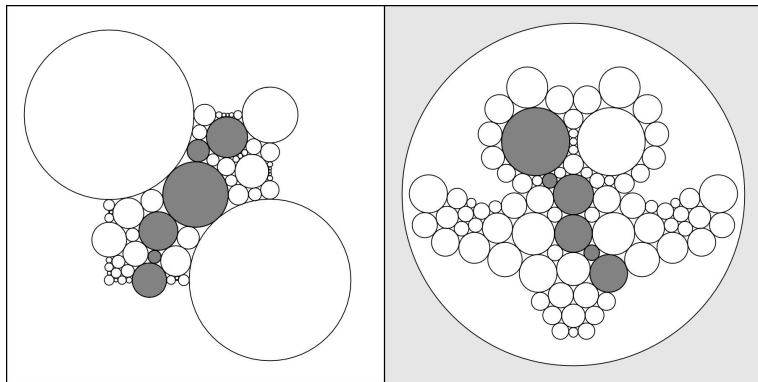
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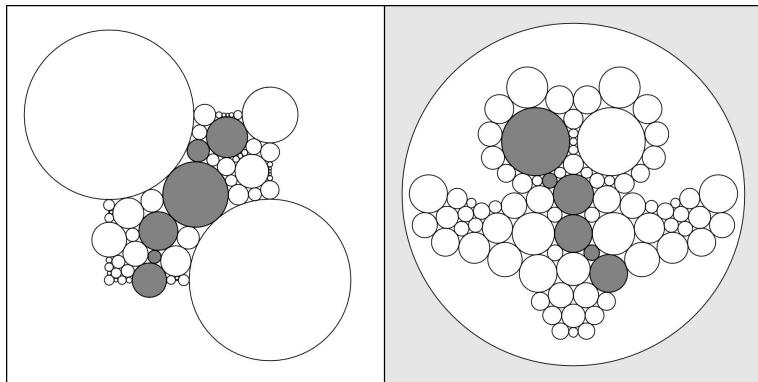
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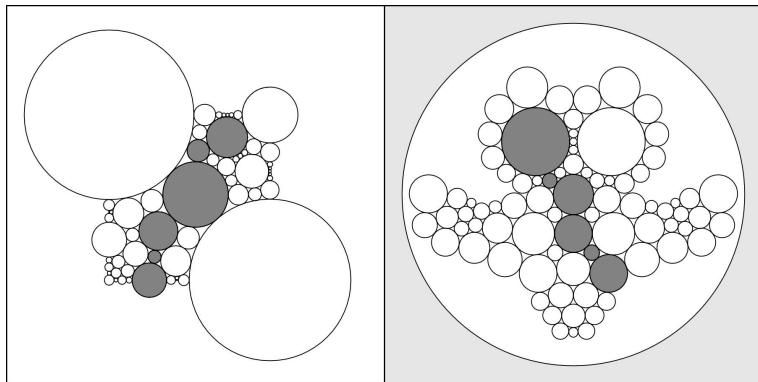
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Analytic functions, *map infinitesimal circles to infinitesimal circles.*

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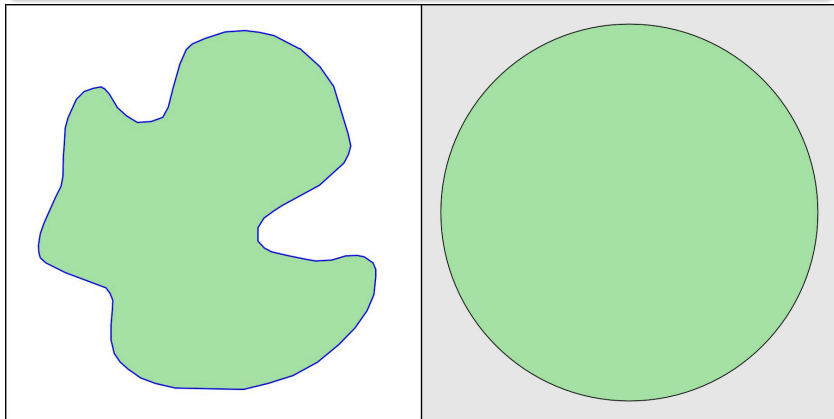


Analytic functions, *map infinitesimal circles to infinitesimal circles.*
Circle packing, *maps real circles to real circles.*

Riemann Mapping Theorem

Classical RMT

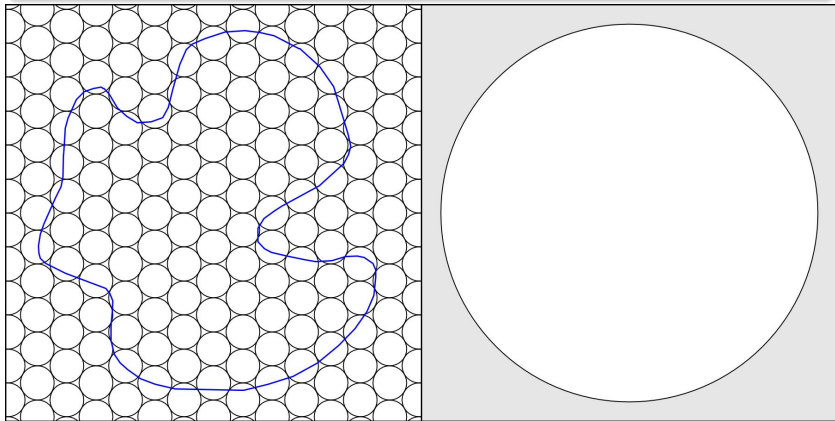
If R is a non-empty simply connected proper subset of $\mathbb{C}$, then there exists an essentially unique mapping from R onto the unit disc.



Riemann Mapping Theorem

Discrete RMT

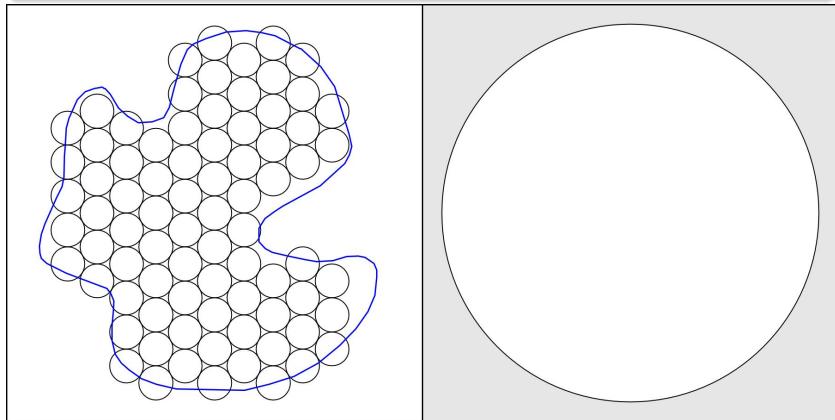
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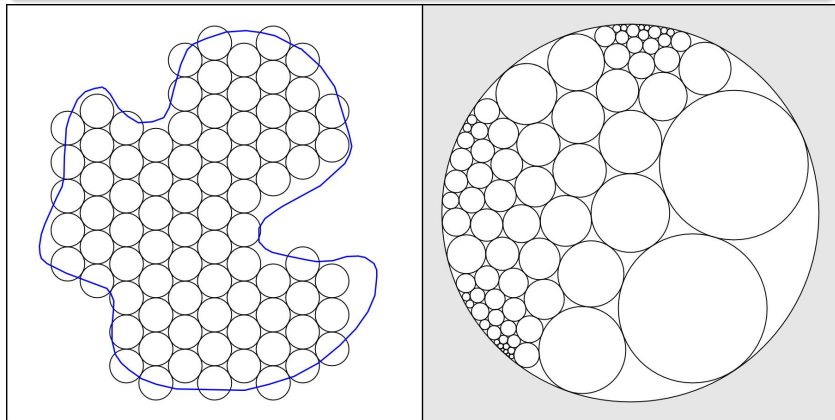
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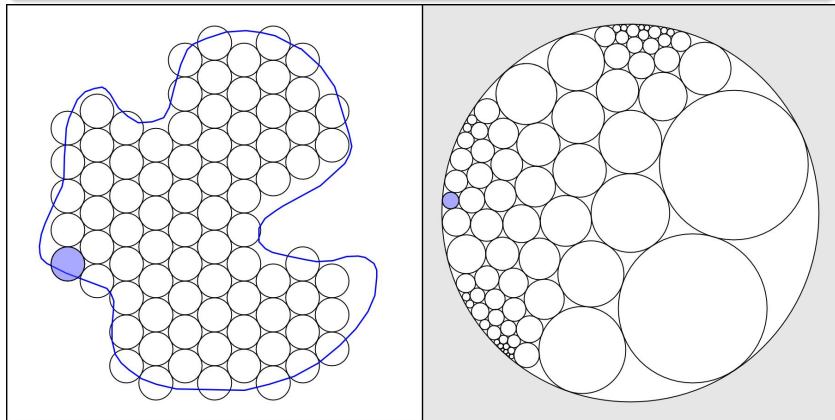
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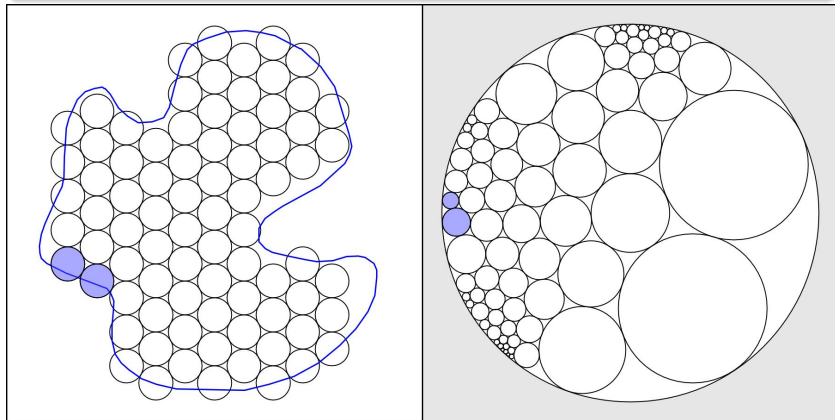
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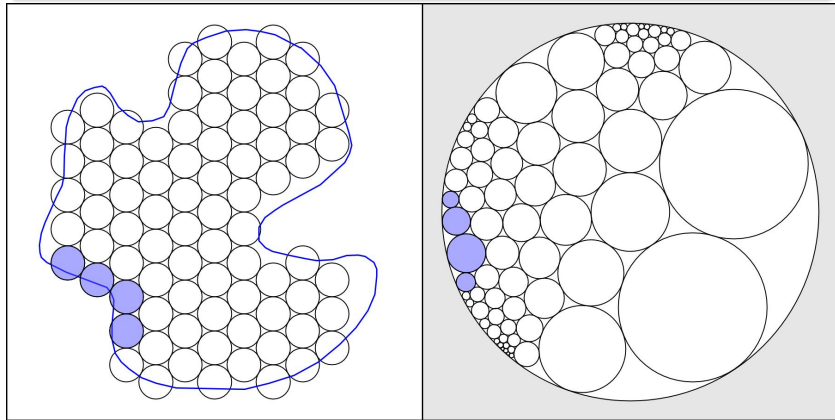
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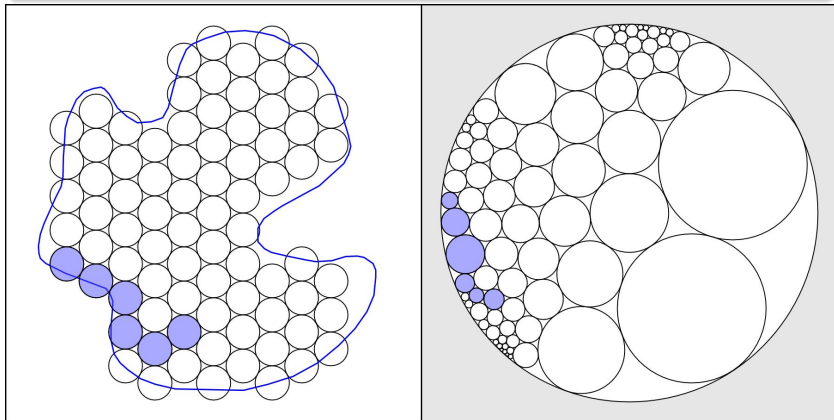
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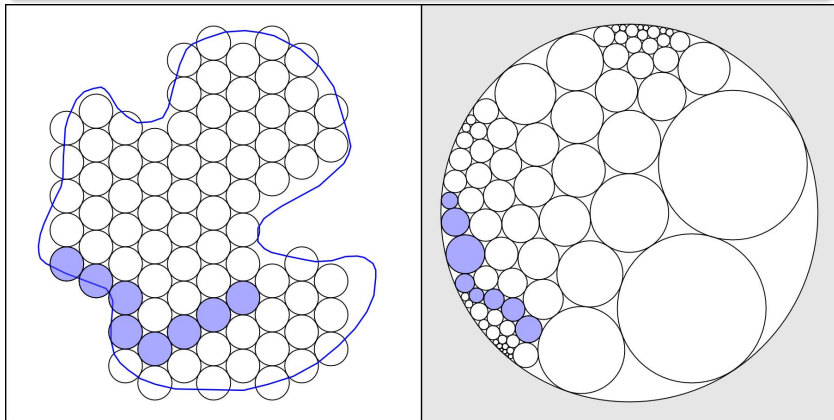
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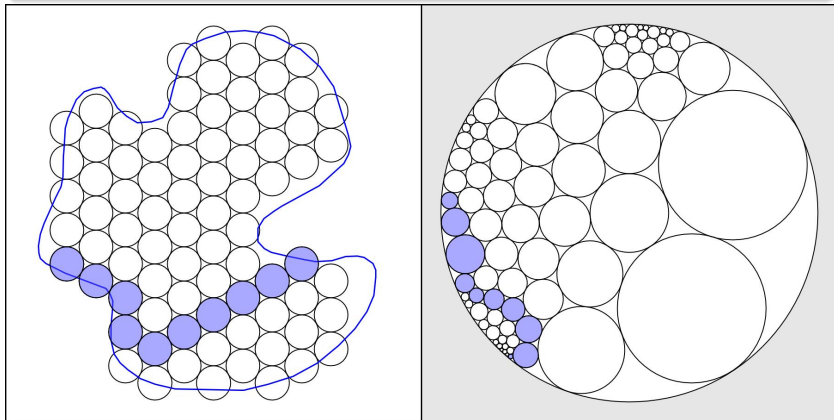
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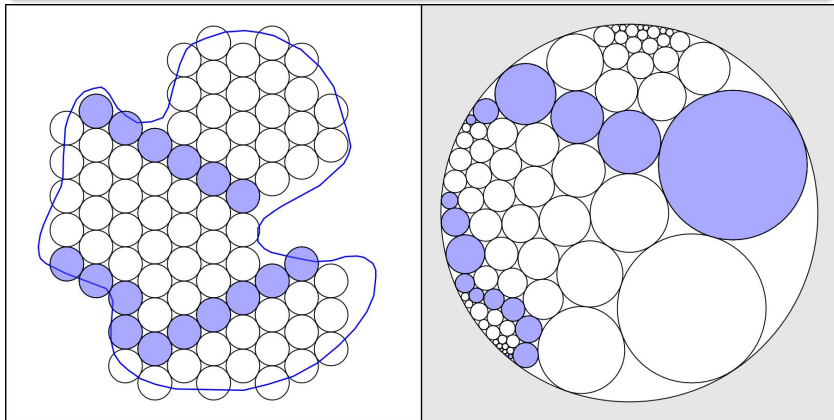
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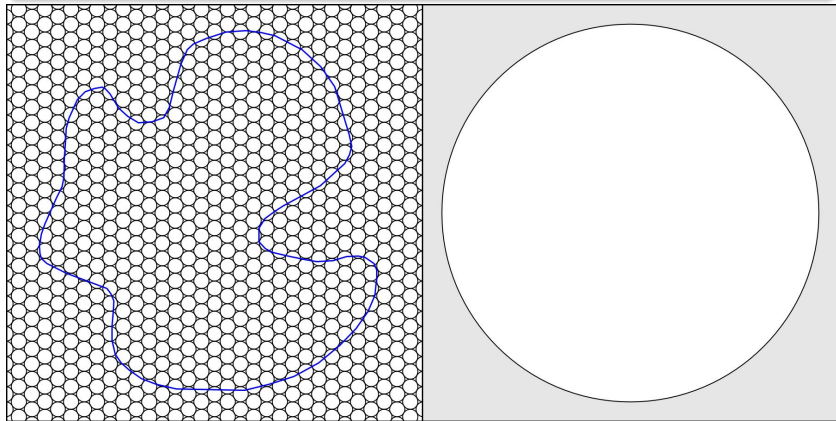
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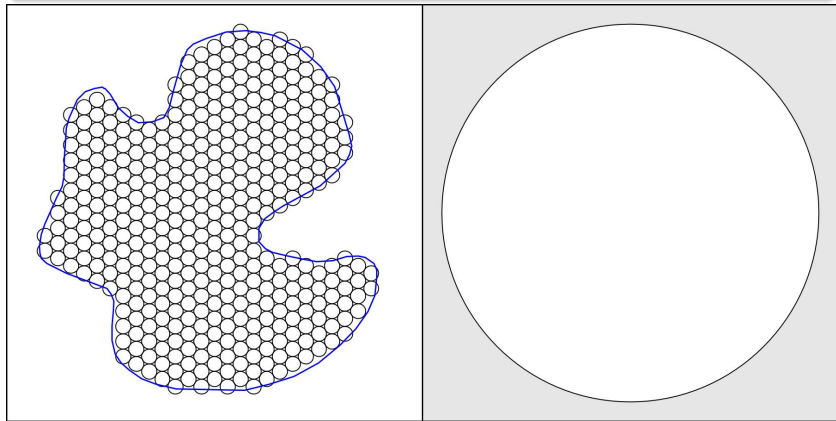
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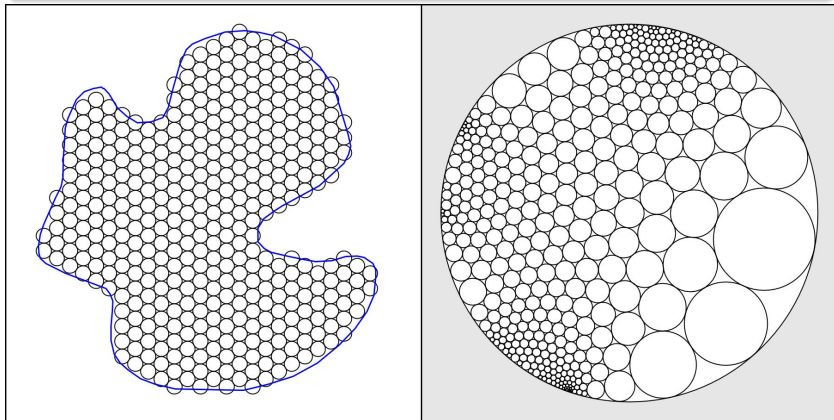
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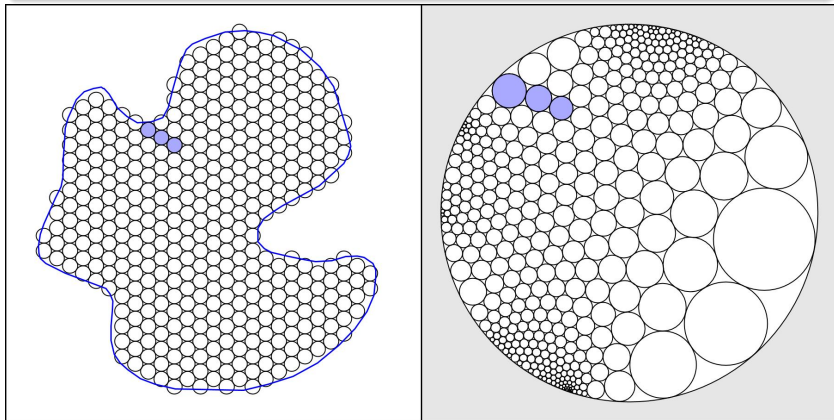
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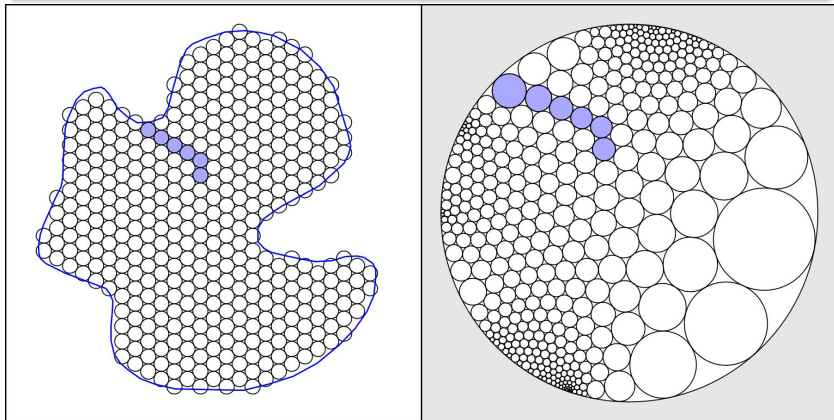
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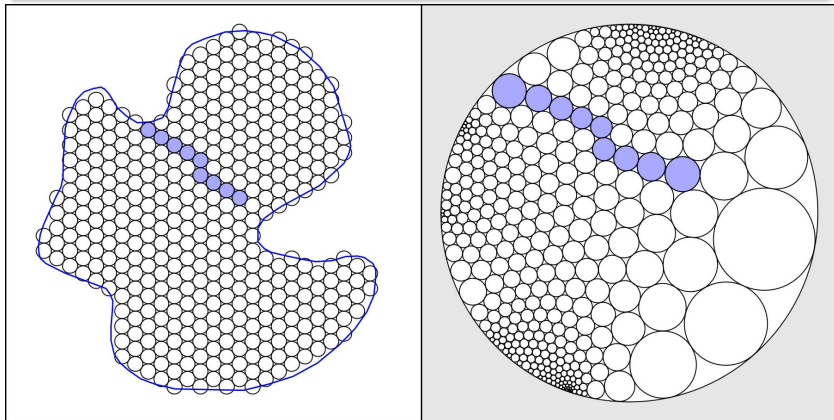
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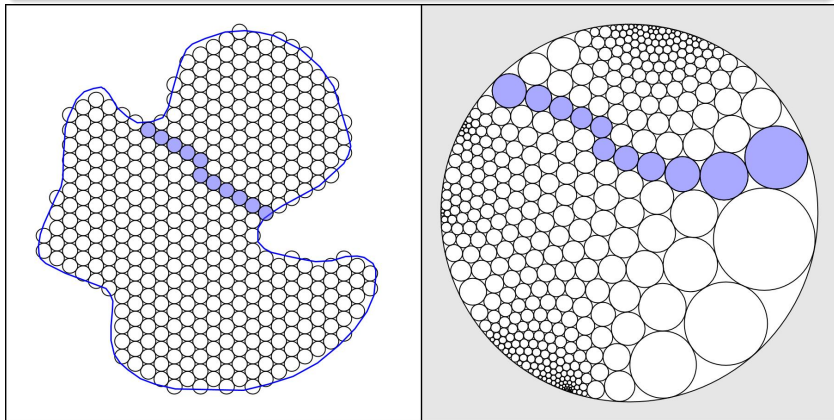
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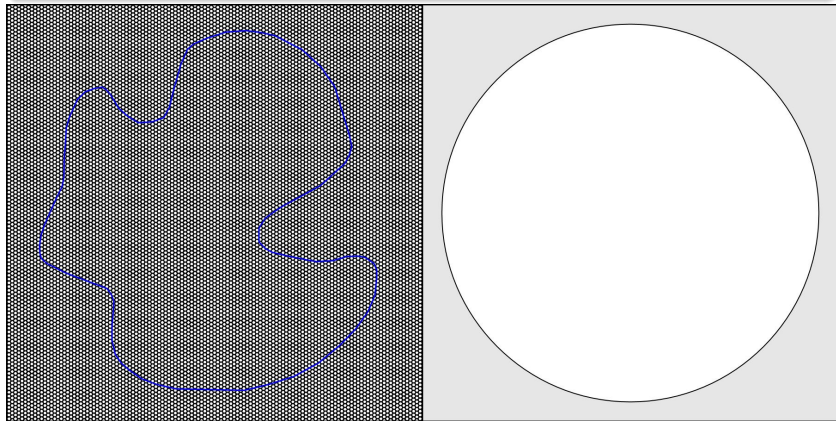
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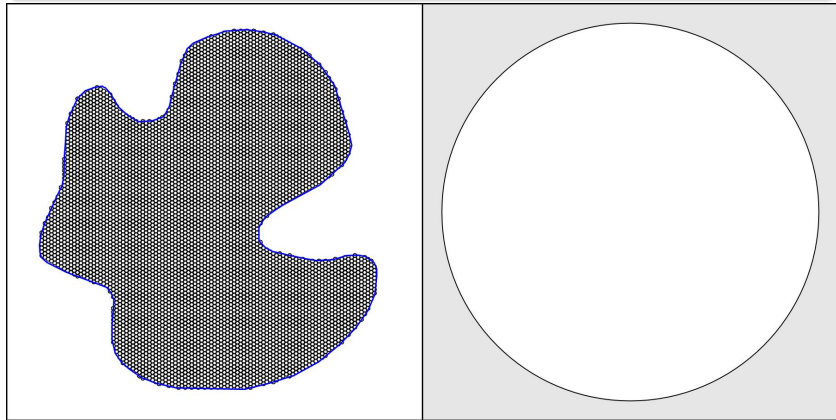
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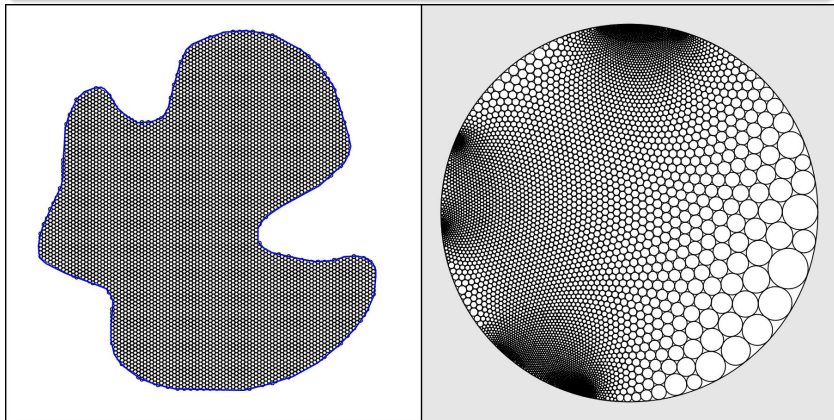
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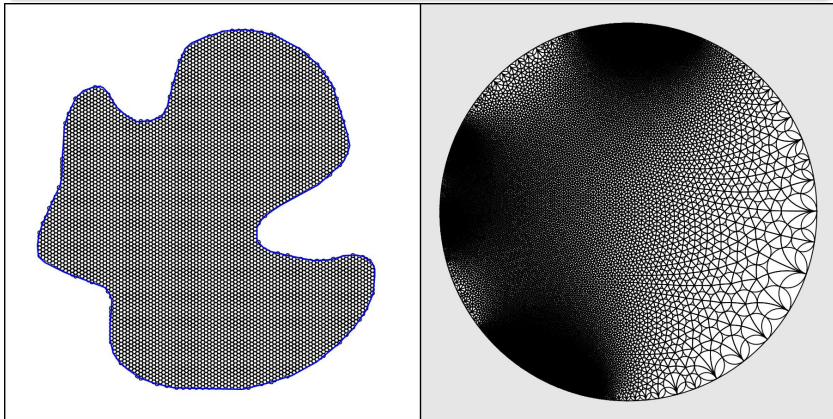
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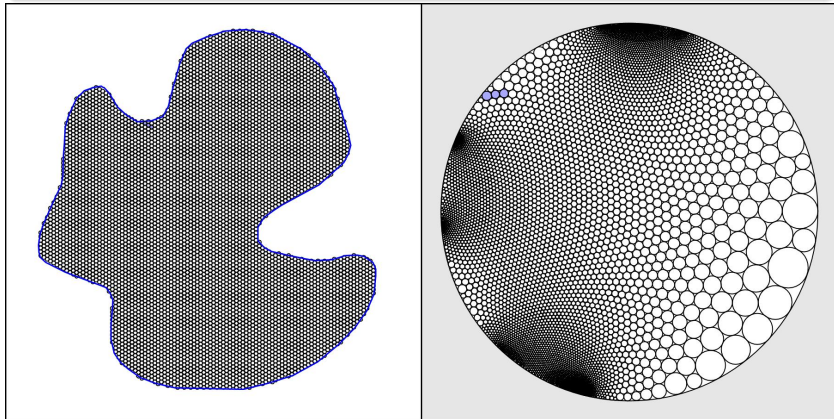
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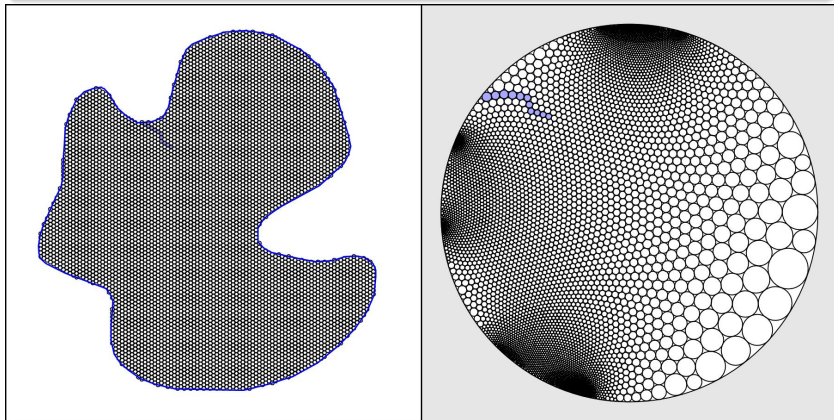
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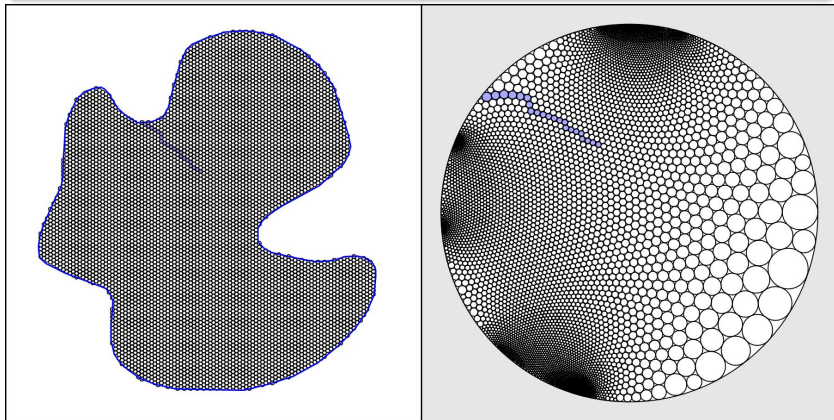
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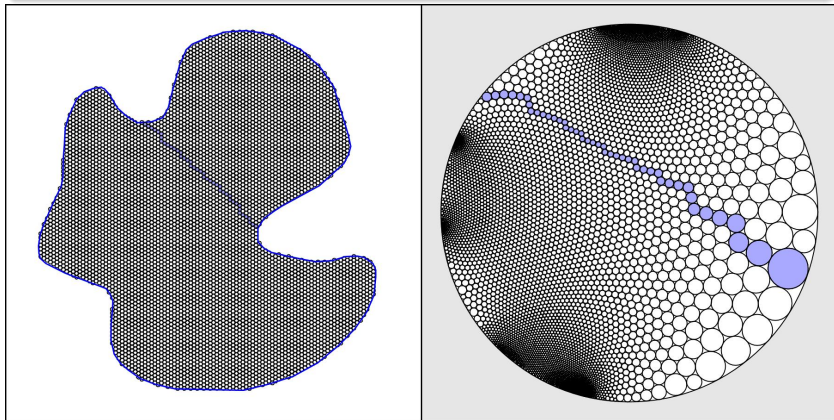
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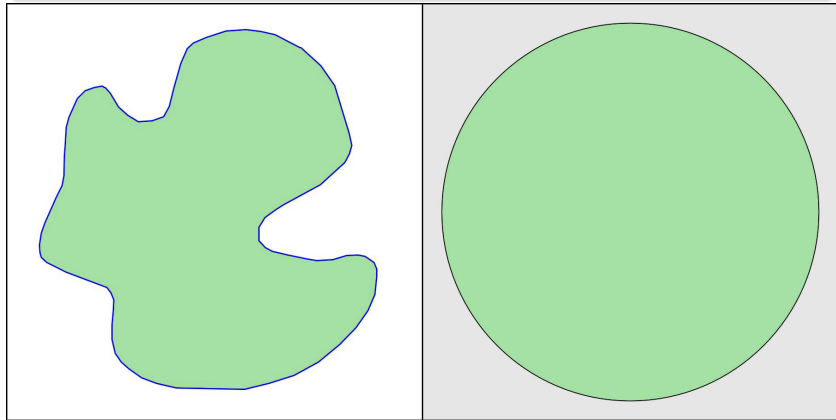


- More circles, and the maps converge to the classical mapping.

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- Discrete Schwarz Lemma

Discrete _____ Function

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- Discrete Blaschke products

Theorem

Discrete analytic functions converge under refinement to their classical counterparts.

Discrete _____ Function

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[He and Schramm] The KAT theorem implies the classical Riemann Mapping Theorem

Discrete _____ Function

- Discrete Schwarz Lemma
- Discrete Blaschke products
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- Discrete rational functions, etc.

Theorem

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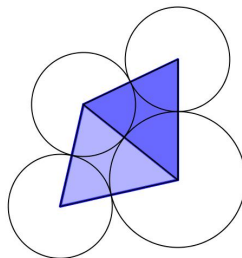
Branching

Flowers with angle sums of

- 2π , a (trivially) branched flower.

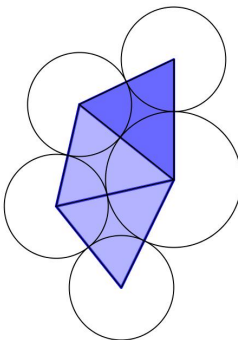
Flowers with angle sums of

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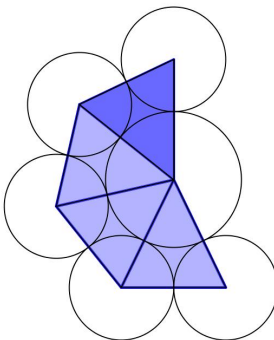
Flowers with angle sums of

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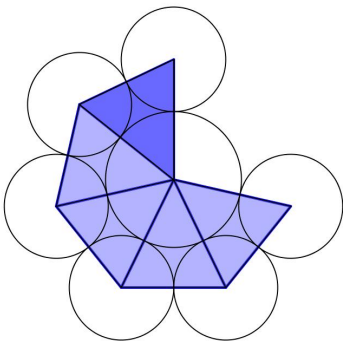
Flowers with angle sums of

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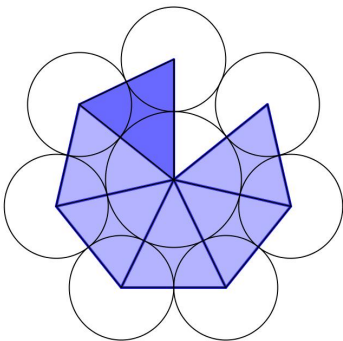
Flowers with angle sums of

- 2π , a (trivially) branched flower.



Flowers with angle sums of

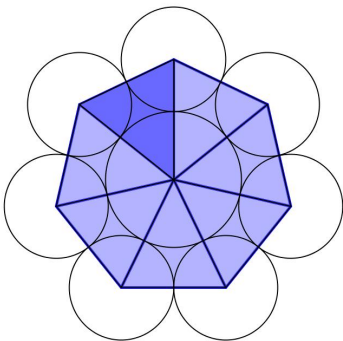
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Branching

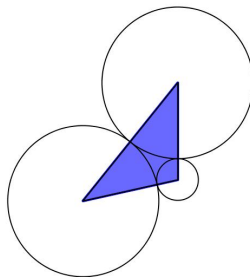
Flowers with angle sums of

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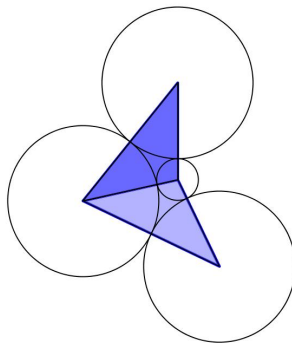
Flowers with angle sums of

- 4π , a branched flower.



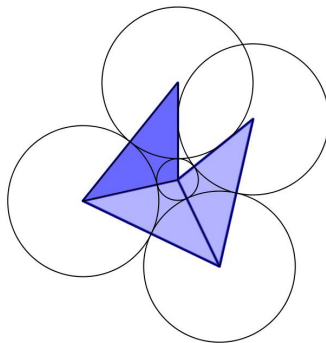
Flowers with angle sums of

- 4π , a branched flower.



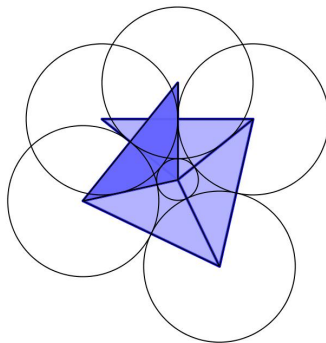
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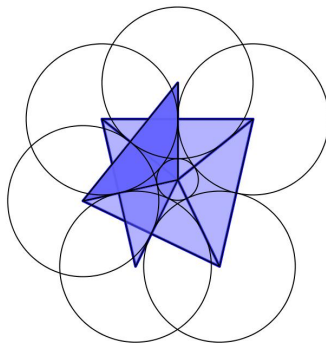
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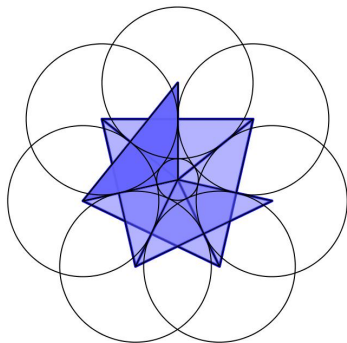
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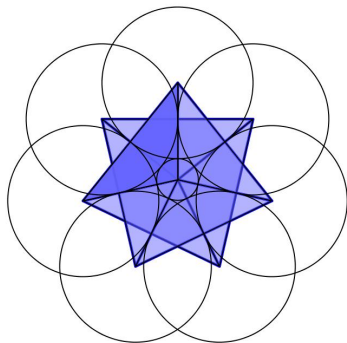
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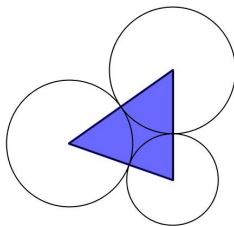
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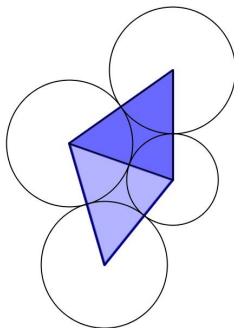
Flowers with angle sums of

- 2.75π , an “incoherent” flower



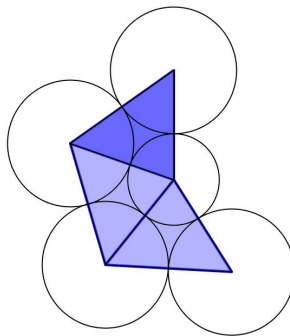
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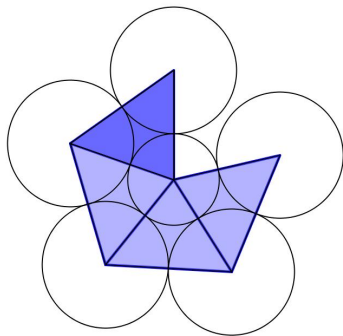
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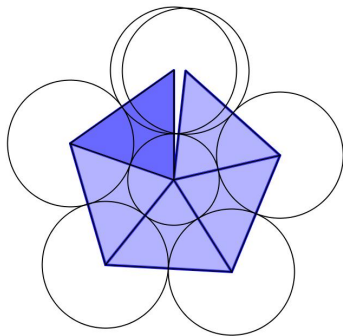
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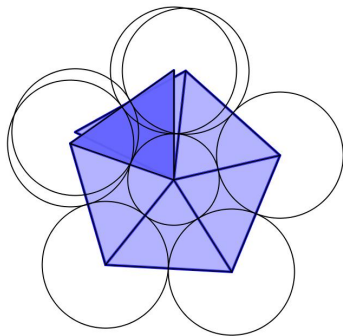
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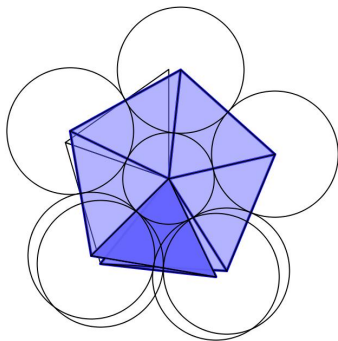
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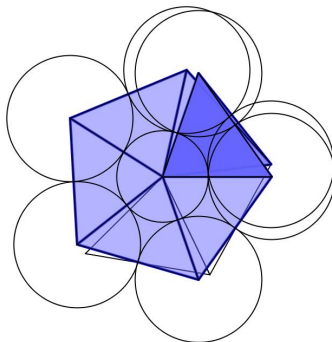
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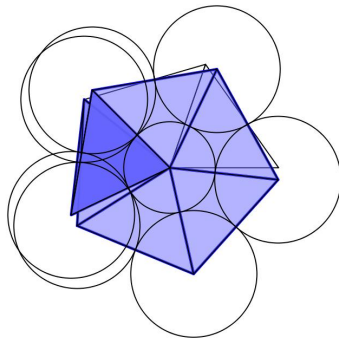
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Branching

Interior flowers with angle sums of

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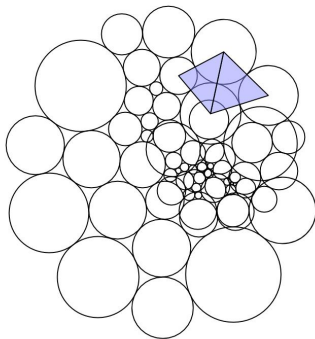
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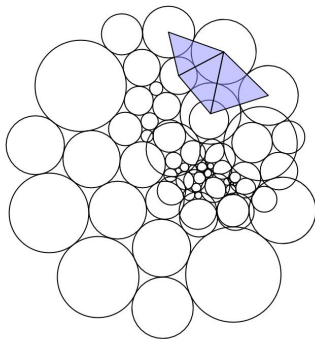
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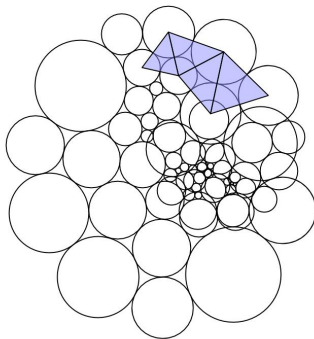
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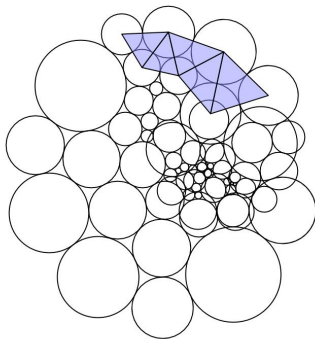
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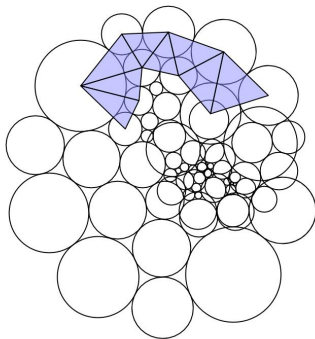
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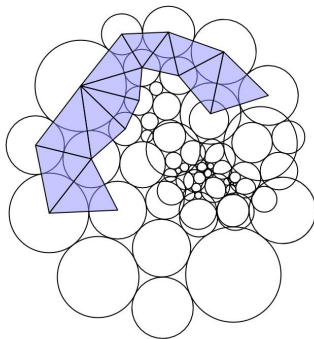
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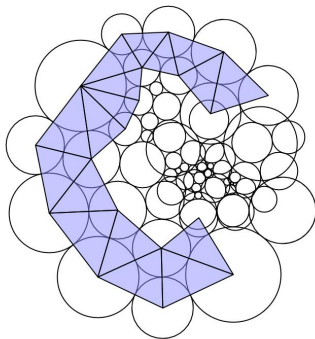
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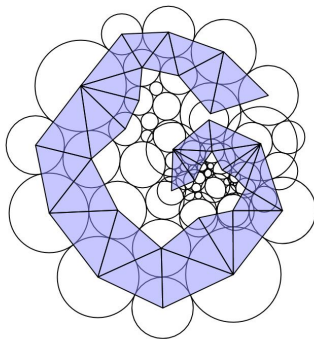
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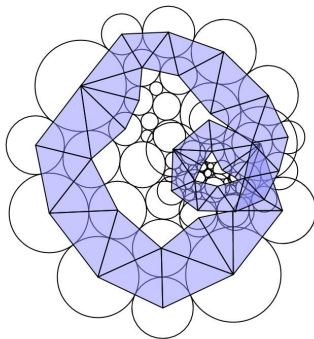
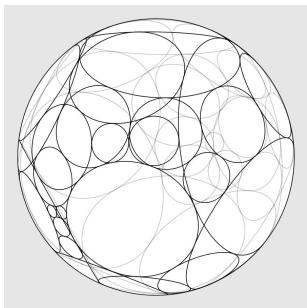
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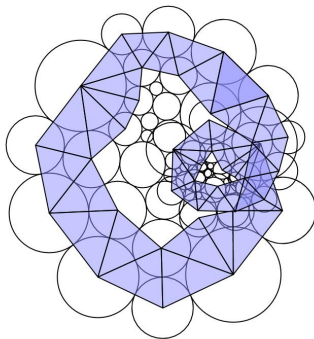
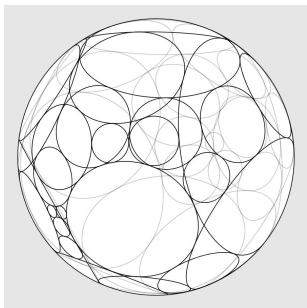


- Given some combinatorial restrictions!

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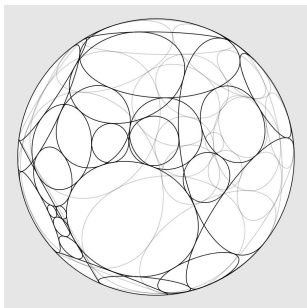


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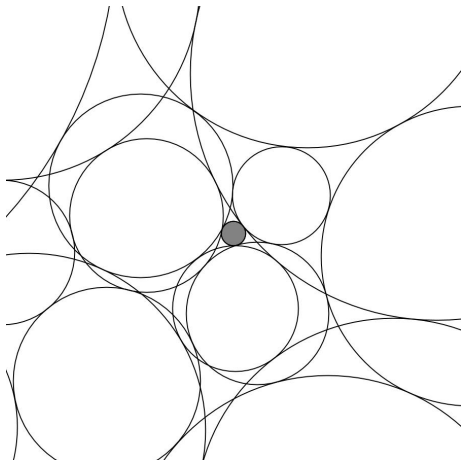
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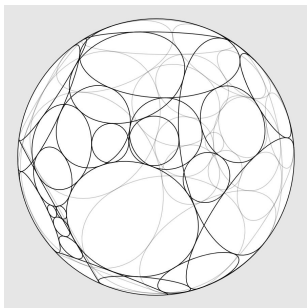
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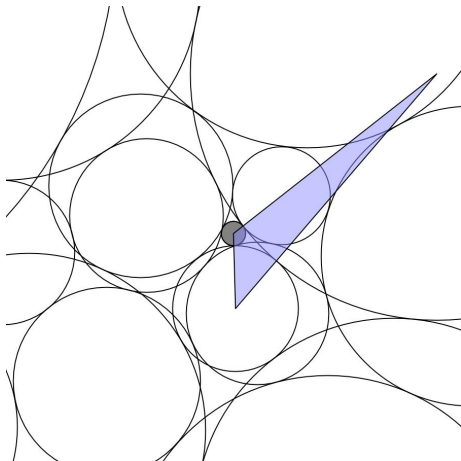
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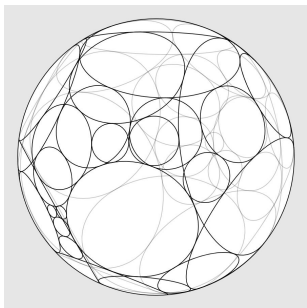
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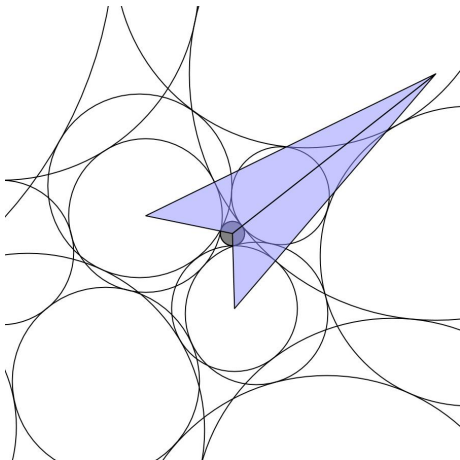
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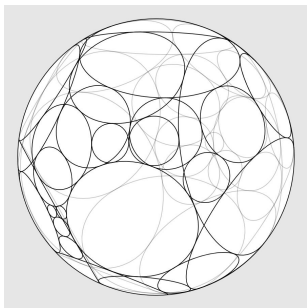
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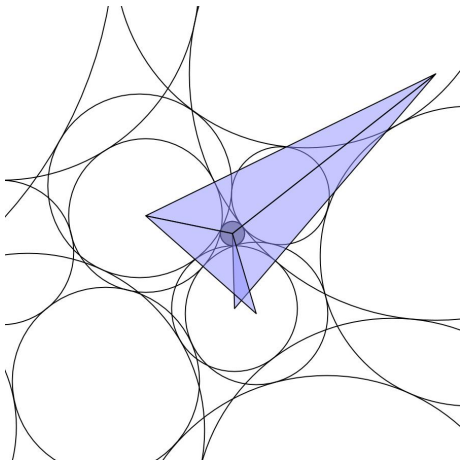
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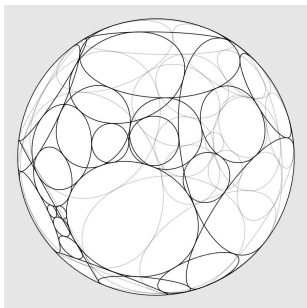
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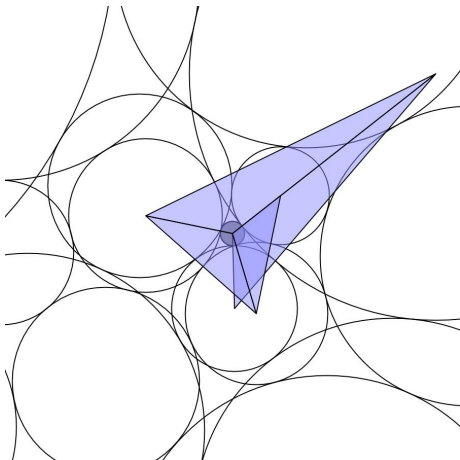
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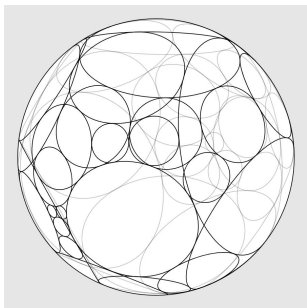
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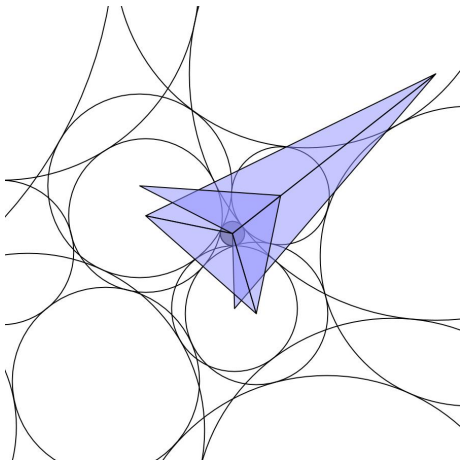
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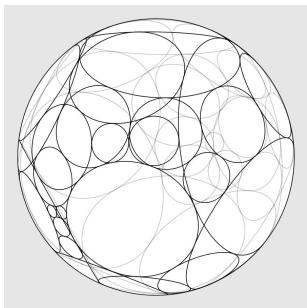
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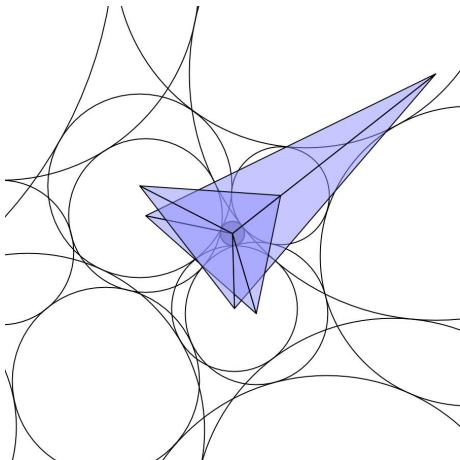
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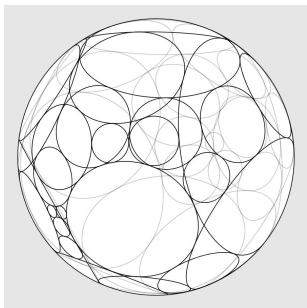
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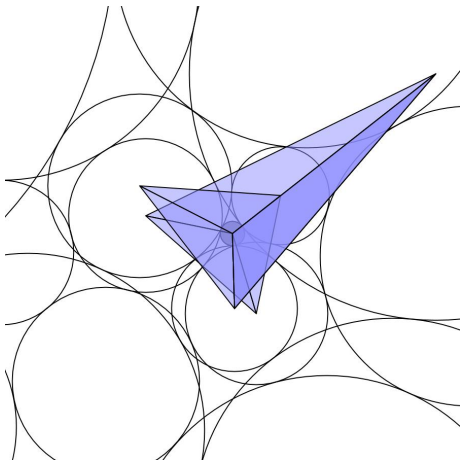
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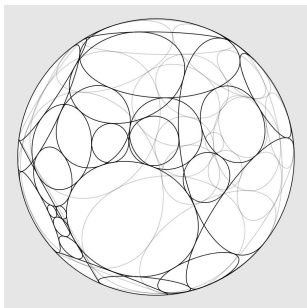
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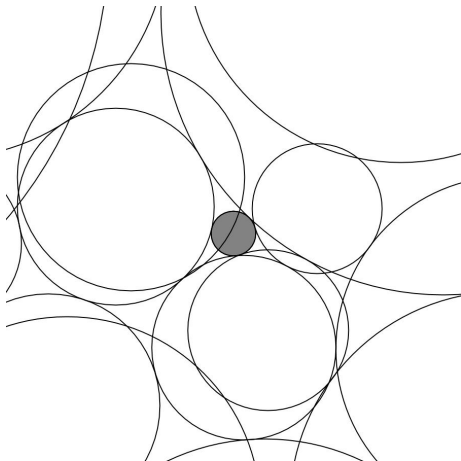
Branching

Interior flowers with angle sums of

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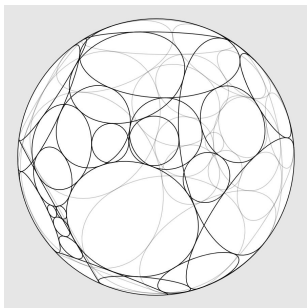
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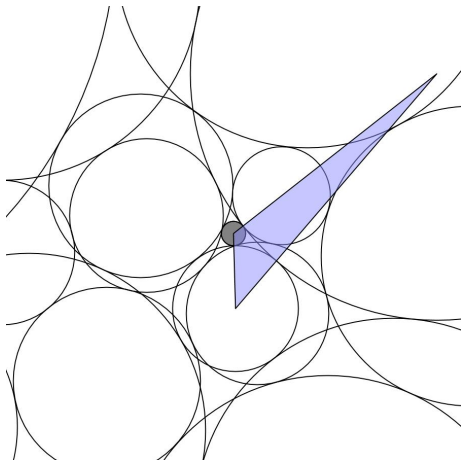
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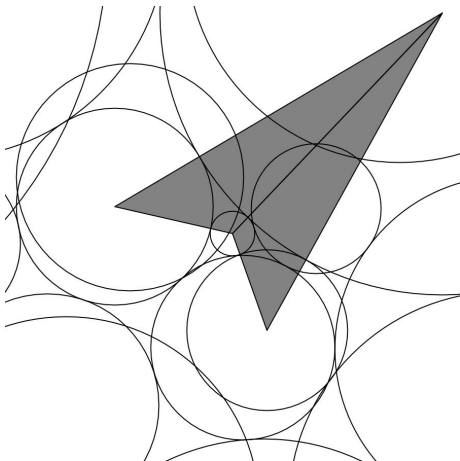
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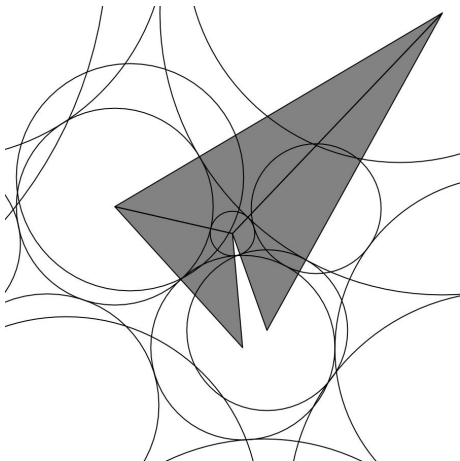
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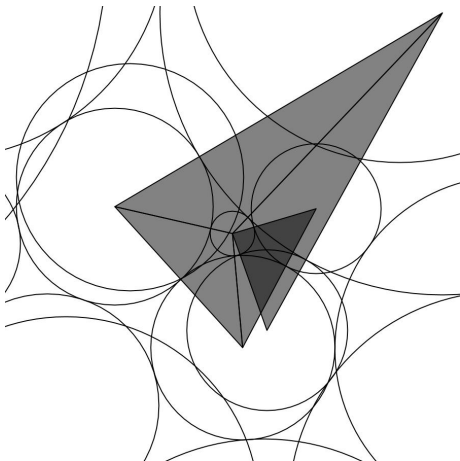
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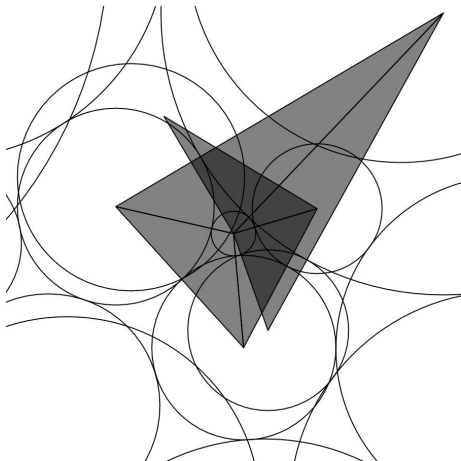
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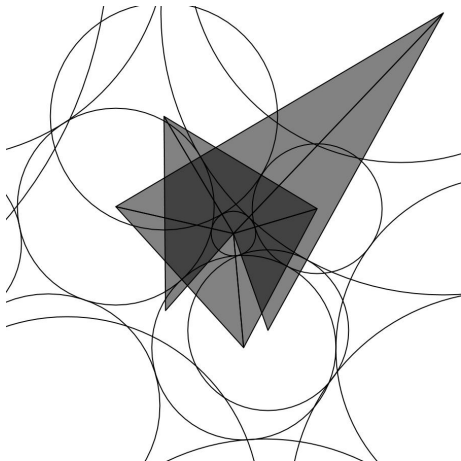
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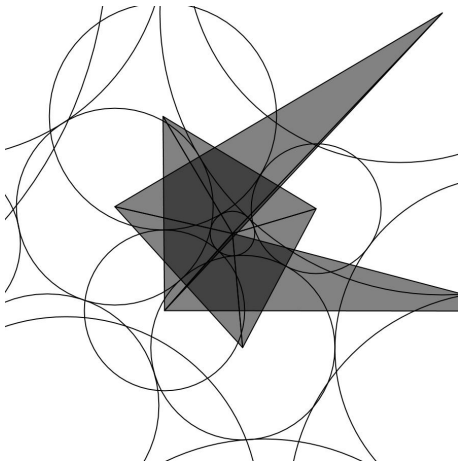
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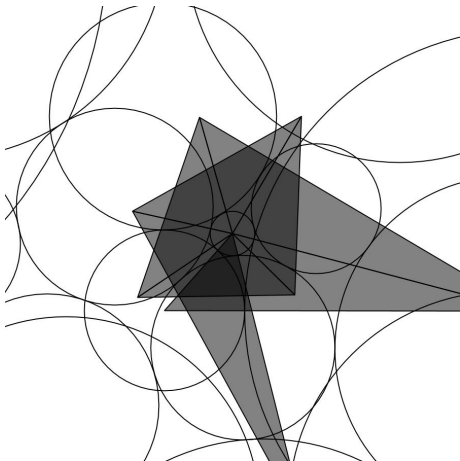
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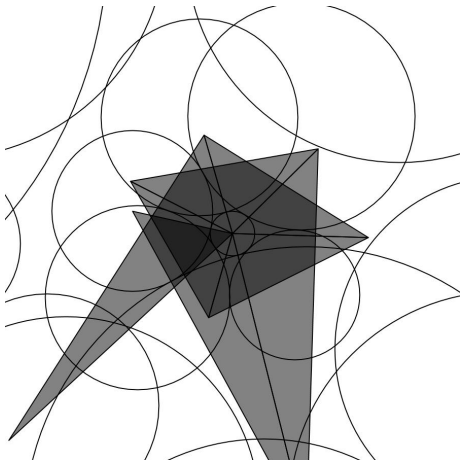
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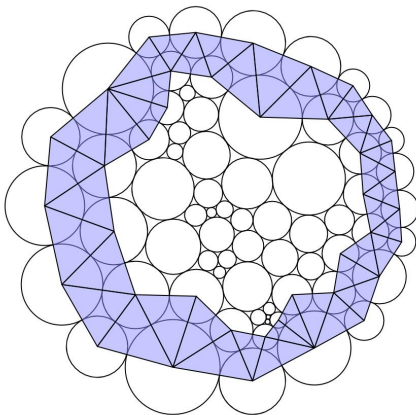
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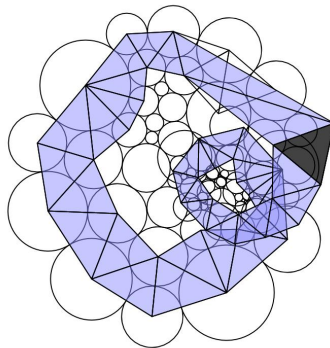
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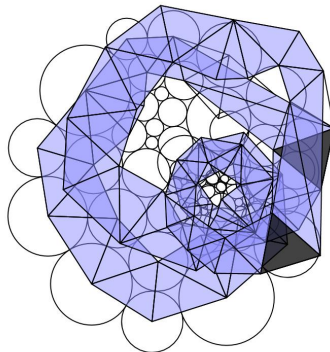
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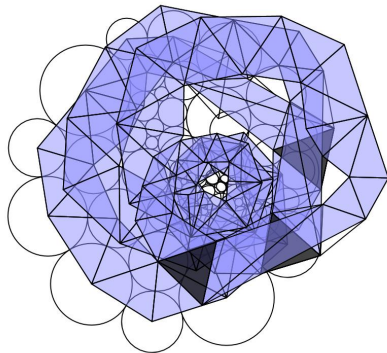
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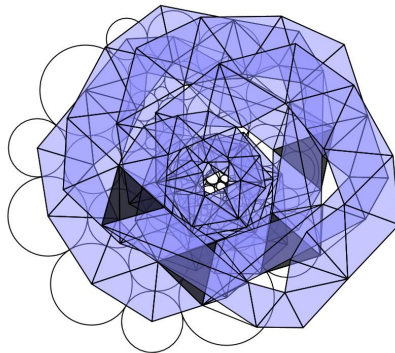
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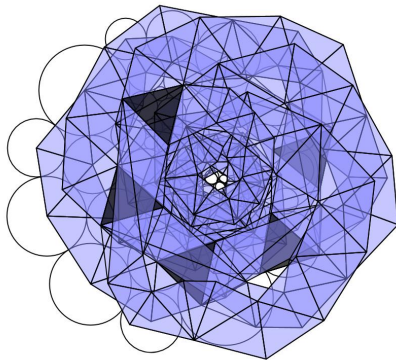
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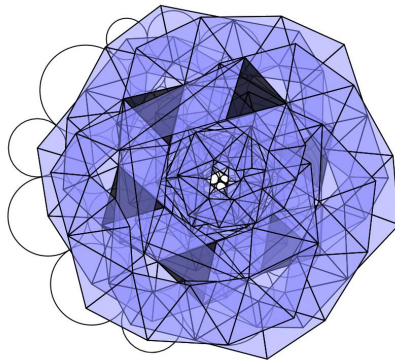
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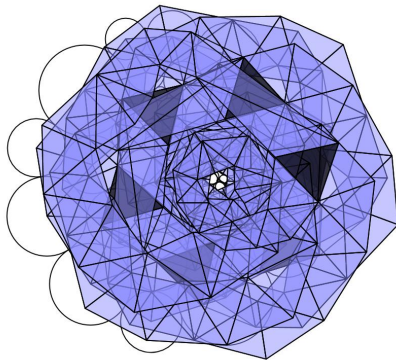
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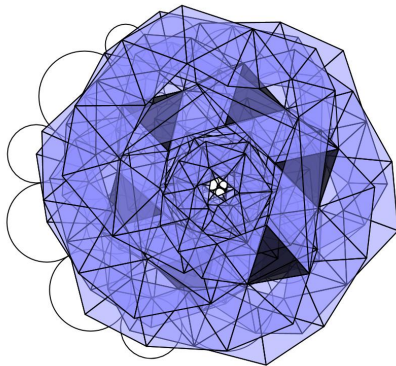
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Fractional Branching

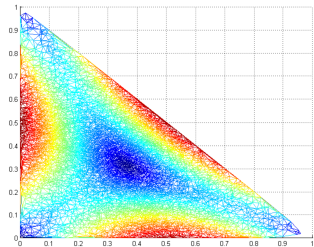
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Is there a way to split the extra angle amongst the vertices to get a branched packing?

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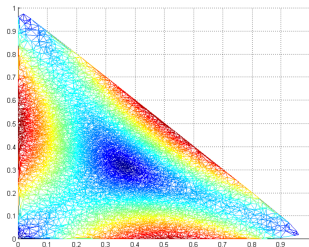


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Theorem

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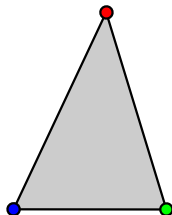


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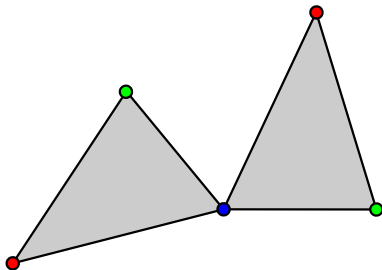


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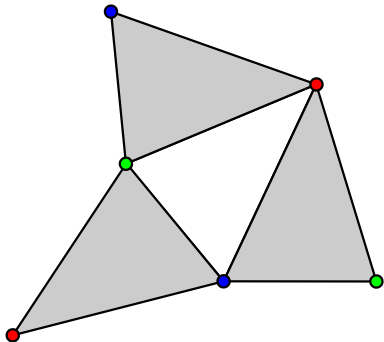


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Choose an interior face $f = \{v_1, v_2, v_3\}$.
Is there a way to split the extra angle
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Theorem

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- Proof using Brouwer fixed point theorem.

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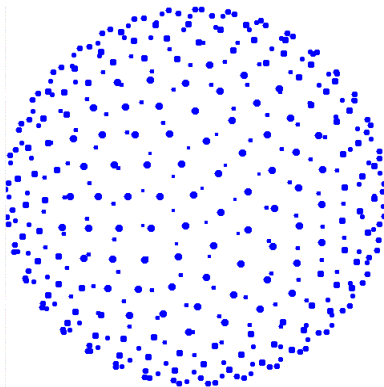
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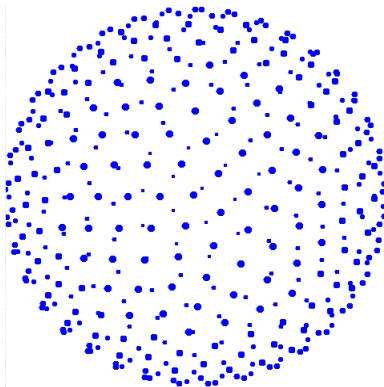


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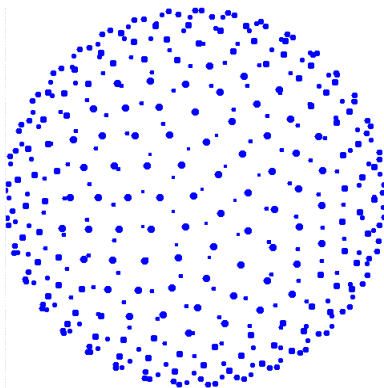
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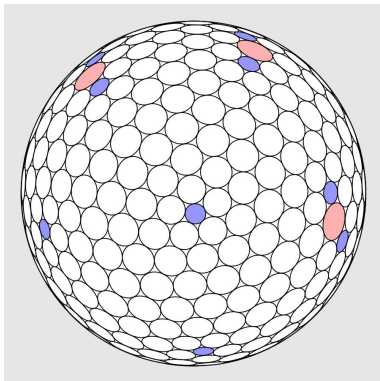
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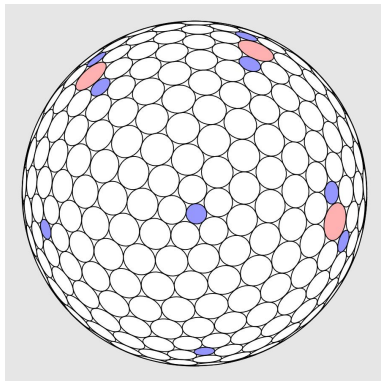
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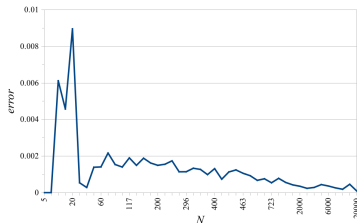


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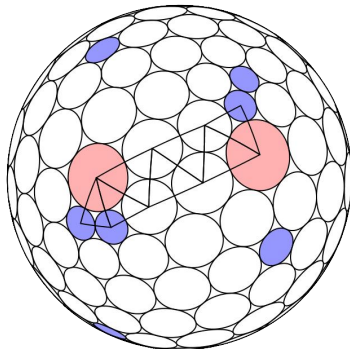
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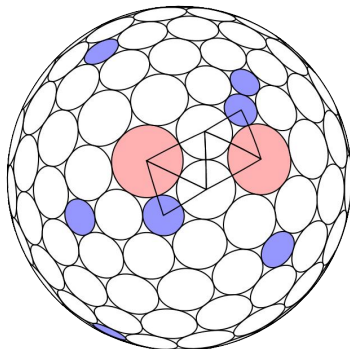
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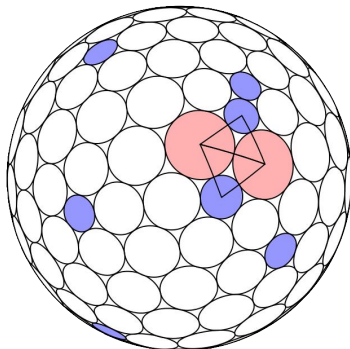
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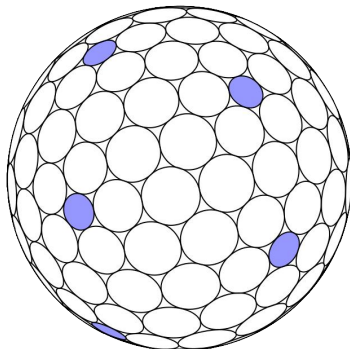
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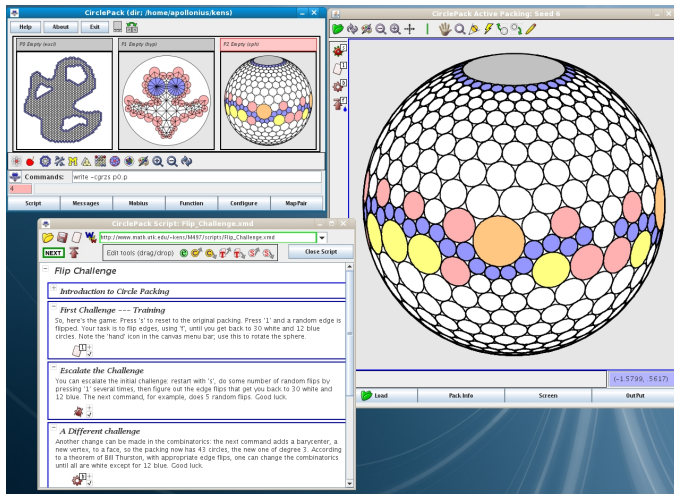
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All images created with CirclePack

<http://www.math.utk.edu/~kens/CirclePack/>



Thanks!